

Fangyu HOU

Applied Physics Researcher

Profile

- Applied physics researcher working across multimodal olfactory sensing, biophysics, functional materials, scientific computing, and human-centered AI.

Education

Mas- ter of Sci- ence	City University of Hong Kong , Applied Physics • Dissertation supervisor: Xiangqiang Chu	Hong Kong, China Sept 2023 – Oct 2024
Bach- elor of Sci- ence	Lanzhou University , Physics • Specialization: Metallic Materials Physics • Undergraduate thesis supervisor: Dongsheng Lei • Outstanding Student League Cadre, 2020 and 2021 (Top 5%)	Lanzhou, China Sept 2019 – July 2023

Experience

The Future Lab, Tsinghua University , Research Assistant Multimodal Olfactory Sensing and Computing • Led a cotton-smoldering early-warning study combining field investigations, AHP-based risk modeling, and TG-FTIR-GC-MS characterization to guide multimodal sensor and experimental-system design. • Optimized thin-film sensing layers and electrode/substrate architectures using spin coating, layer-by-layer assembly, and ultrasonic spray coating. • Improved room-temperature NH ₃ sensing to reproducible cycles at 10-25 ppm and achieved measurable responses to H ₂ S and CH ₃ SH. • Co-developed E-Nose Station for response-curve visualization, cross-experiment comparison, odor-fingerprint analysis, model-assisted screening, and semi-automated acquisition workflows.	Beijing, China Jan 2026 – present 10 months
Great Bay University , Research Assistant Biomimetic Colored Radiative-Cooling Films • Fabricated and characterized microstructured colored radiative-cooling films using vacuum deposition, UV-Vis spectroscopy, infrared and thermal imaging, nanoindentation, and microscopy. • Processed spectral and imaging data to analyze relationships among microstructure, visible appearance, optical transparency, and thermal behavior.	Dongguan, China Sept 2025 – Dec 2025 4 months
Institute of Physics, Chinese Academy of Sciences , Research Assistant GSDM Protein Dynamics and Cell Pyroptosis • Used HS-AFM to visualize pore-forming dynamics of ThGSDM, GSDMD, and GSDME on bare mica and DOPC/DOPS supported lipid bilayers. • Contributed to X-ray crystallography data processing.	Beijing, China Sept 2024 – Dec 2024 4 months

National Undergraduate Innovation and Entrepreneurship Training Program, Undergraduate Researcher
Flexible-Substrate Metasurface Optical Tweezers

- Designed PDMS flexible-substrate metasurface lenses and nanopillar unit cells using FDTD simulations and MATLAB optical-force modeling.
- Modeled substrate deformation and demonstrated stable simulated trapping of 15 nm polystyrene spheres near the self-focusing region.

Lanzhou, China
Mar 2022 – Mar 2023
1 year 1 month

Projects

Locus: Human-AI Decision Support for PhD Research Fit and Advisor Discovery

Independent project

- Designed and developed a human-AI decision-support platform for personalized PhD research positioning and advisor exploration.
- Built an adaptive advisor search and ranking engine using offline faculty retrieval, LLM-generated candidates, research-similarity scoring, and institution and discipline constraints.
- Added uncertainty-aware verification using web, OpenAlex, and ORCID evidence and piloted the platform with more than 50 users.

Beijing, China
Apr 2026 – present

Publications

Computational Design of 3D-Printable Freeform Caustic Surfaces

Shuowen Li, Danqi Huang, *Fangyu Hou*, Qiuyu Lu, Haipeng Mi, Zhihao Yao

[10.1145/3799825.3818720](https://doi.org/10.1145/3799825.3818720) (ACM SIGGRAPH 2026 Posters, Article 4)

Jan 2026

Patents

1. Cotton Smoldering Simulation and Multimodal Gas Detection Device - Chinese invention patent application, pending (2026)

Skills

Programming and Data Analysis: Python (NumPy, Pandas, Matplotlib, SciPy), MATLAB, SQLite

Web Development and Deployment: JavaScript, HTML/CSS, Flask, Linux, Git, Nginx, web deployment, DNS, SSL/TLS configuration

Scientific Computing and Engineering: FDTD Solutions, COMSOL, SolidWorks, Arduino

Sensing and Experimental Methods: HS-AFM/AFM, spin coating, ultrasonic spray coating, layer-by-layer assembly, and optical, thermal, and mechanical materials characterization